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A co-training approach integrating CNN and Mamba for semi-supervised 3D medical image segmentation

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ABSTRACT

Accurate 3D medical image segmentation is vital for clinical applications but hindered by the scarcity of expert-annotated data, posing challenges for fully supervised models that require extensive labels. To address this issue, semi-supervised learning (SSL) has emerged as an effective solution by enabling models to learn from limited labeled data and achieve performance comparable to fully supervised learning, thereby significantly reducing the burden of manual annotation. Among various SSL strategies, deep co-training has demonstrated its effectiveness, yet current methods suffer from excessive information sharing between subnetworks or convergence during the training process. In this paper, we propose a semi-supervised 3D medical image segmentation method based on heterogeneous co-training, which combines the local feature perception capability of Convolutional Neural Networks (CNNs) with the efficient sequence modeling ability of the Mamba architecture. This integration enables more robust and complementary feature learning. To further enhance the utilization of unlabeled data, we introduce a Confidence-Aware Consistency(CAC) loss, which enforces consistency between model predictions while maintaining the learning capacity of individual networks. In addition, we propose a TriMix data augmentation strategy and incorporate a feature perturbation mechanism to increase the diversity of training samples and improve generalization. Extensive experiments were performed on three publicly available datasets: BraTS2019, left atrium (LA) and pancreas. The proposed method achieved superior performance compared to existing semi-supervised segmentation approaches, as measured by Dice score and HD95, demonstrating its effectiveness in enhancing the accuracy and robustness of 3D medical image segmentation with limited labeled data.

1. Introduction

Medical image segmentation is a critical task in medical image analysis, aiming to accurately delineate structures and lesion regions of interest, thereby playing a vital role in assisting diagnosis and treatment planning. In recent years, fully supervised methods based on deep learning have achieved remarkable performance in medical image segmentation [1]. Nevertheless, conventional supervised learning-based segmentation methods face several challenges. First, the success of traditional approaches heavily relies on large volumes of high-quality annotated data for training. Due to the complexity and domain-specific nature of medical images, annotations must be performed by medical experts with extensive clinical experience, making the process labor-intensive, time-consuming, and demanding high levels of expertise. Additionally, medical images often contain sensitive information involving patient privacy, and strict regulations governing data acquisition and usage further limit the availability of training data.

In recent years, to mitigate the dependency on large-scale labeled datasets, semi-supervised learning (SSL) methods have garnered significant attention. These methods leverage a small amount of labeled data alongside a large pool of unlabeled data, substantially reducing annotation costs [2]. In the field of semi-supervised segmentation, researchers have primarily explored techniques such as consistency regularization, pseudo-labeling, and self-training to improve model performance. However, for 3D medical image segmentation, traditional SSL frameworks often struggle to fully exploit the rich spatial and contextual information inherent in volumetric data. The mean teacher [3] and co-training [4] paradigms are among the most widely

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adopted frameworks in current research. However, the teacher–student scheme often causes both networks to converge to similar local minima, thereby reducing the effect of perturbations and ultimately impairing the model's generalization ability [5]. In addition, some challenging medical segmentation cases may be consistently segmented by both the teacher and student models, which can introduce confirmation bias during training. These limitations motivate us to introduce the co-training framework. Co-training involves training two sub-networks to infer the same instance from different perspectives, enabling them to provide diverse and complementary information to each other. This helps reduce confirmation bias and improves prediction accuracy. Cross pseudo supervision(CPS) [6] was the first method to apply co-training to the problem of semi-supervised semantic segmentation. However, when multiple sub-networks are used to process data collaboratively, excessive information sharing or overly similar learning processes may lead to their convergence in feature extraction or prediction, thereby diminishing the ability to learn from distinct perspectives. This convergence can be viewed as a form of collapse, as it undermines the overall diversity and robustness of the model. A key challenge in co-training is to develop strategies that mitigate the risk of mutual collapse between sub-networks. Employing different input domains or heterogeneous architectures can effectively reduce prediction similarity and yield richer complementary information.

Various improvements have been proposed to address the limitations of co-training in semi-supervised semantic segmentation. These methods can be categorized into three groups based on their primary strategy: data-level, feature-level, and network-level [7]. Each category introduces model diversity from a different perspective to mitigate pseudo-label noise and confirmation bias. However, these methods still face challenges in controlling the quality of pseudo labels. It is also essential to ensure that, while the two sub-networks are encouraged to learn diverse yet complementary features, unreliable outputs are prevented. To address this issue, Wang et al. [8] proposed the conflict-based pseudo-labeling, which allows the sub-networks to extract richer semantic information from conflicting predictions and ultimately achieve consistent and reliable outputs. Such strategies underscore the importance of pseudo-label selection mechanisms, suggesting that the design of more effective co-training schemes and robust pseudo-label selection strategies remains a critical research direction.

In previous studies on fully supervised methods, CNNs have significantly improved the accuracy of 3D medical image segmentation owing to their strong feature extraction capabilities. However, the local convolution operations of CNNs are limited in capturing long-range dependencies. The Transformer architecture alleviates this issue by modeling long-range dependencies, but its computational complexity grows quadratically with input resolution, posing challenges to efficiency. Recently, the Mamba architecture, based on state-space models (SSMs), has been proposed as an efficient alternative to traditional Transformers and has garnered significant attention. Traditional SSMs [9–12] have been successful in domains involving signal processing and time series analysis, but are not applicable to medical image analysis. Mamba is capable of modeling long-range dependencies and performing efficient sequence learning [13]. Compared to Transformers, it offers lower memory consumption and linear-time inference complexity, making it particularly suitable for high-resolution 3D medical image analysis tasks. This opens up new possibilities for building lightweight and robust semi-supervised co-training frameworks.

To address the challenges of limited labeled data, insufficient modeling of long-range dependencies, and confirmation bias in co-training, as illustrated in Fig. 1, we propose a semi-supervised segmentation framework that integrates two structurally heterogeneous networks: one branch adopts a CNN to capture local details, while the other utilizes Mamba for efficient global context modeling. The two branches are jointly trained using a cross pseudo supervision strategy, allowing each to refine its predictions by learning from the other. In order to alleviate the problem of noisy or unreliable pseudo labels, we further

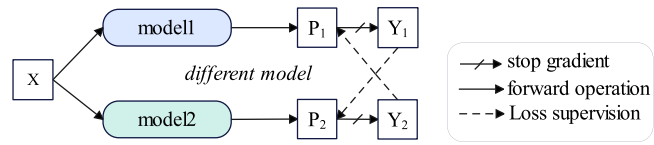

Fig. 1. Illustrating the basic architecture for Ours.

introduce a confidence-aware consistency (CAC) loss that modulates the supervision strength based on prediction confidence, thus improving training stability and pseudo-label reliability. Furthermore, we design a 3D data augmentation strategy named TriMix, and introduce an additional feature-level perturbation method, both of which enhance robustness and improve generalization to complex anatomical variations.

In summary, our main contributions are as follows:

(1) We propose a co-training framework that combines the local modeling capacity of CNNs with the global representation ability of Mamba, enabling complementary feature learning and mitigating representational collapse under limited supervision.

(2) We introduce a CAC loss that adaptively adjusts supervision strength according to pseudo-label confidence, thereby stabilizing training, improving the utilization of unlabeled data, and mitigating confirmation bias inherent in co-training.

(3) Our augmentation strategy integrates two distinct approaches, namely TriMix and a feature-space perturbation module, to enhance model robustness and generalization.

The remainder of this paper is organized as follows: Section 2 reviews related work on medical image segmentation and semi-supervised learning. Section 3 presents the detailed methodology of our proposed approach. Section 4 introduces the experimental datasets and evaluation metrics, presents the ablation and comparative experiments, and provides an analysis of the results. Section 5 describes the computational analysis and future work of the overall proposed method. Finally, Section 6 concludes the paper.

2. Related works

2.1. Fully supervised medical image segmentation

Fully supervised learning has long been the dominant paradigm in medical image segmentation, owing to its ability to leverage rich pixel- or voxel-level annotations to train powerful deep models. This paradigm has witnessed rapid development with the advent of CNNs, vision transformers and Mamba.

Segmentation methods based on CNNs can implicitly learn and extract abstract features from images through deeper network architectures, enabling the modeling of more complex image data representations. The emergence of deep learning-based segmentation algorithms can be traced back to the Fully Convolutional Network (FCN) [14]. Ronneberger et al. [15] proposed the U-Net, the most fundamental segmentation network in medical image segmentation, which adopts a classic encoder–decoder structure with skip connections to fuse low-level and high-level semantic features, thereby significantly enhancing segmentation accuracy. Ozan Oktay et al. [16] introduced attention mechanisms into U-Net, which strengthens the focus on specific features that contribute to better segmentation performance while suppressing irrelevant features, thereby enabling the network to concentrate more on the target task. Özgün Çiçek et al. [17] developed 3D U-Net, an extension of U-Net designed for 3D image segmentation. 3D U-Net employs 3D convolutions to accurately segment volumetric medical images. For 3D medical images, Fausto et al. [18] proposed the V-Net architecture, which replaces the pooling layers in the encoder and decoder with convolutional layers to better preserve and utilize spatial information.

Transformer [19] has been widely adopted in the image processing field due to its powerful global modeling capability. Hatamizadeh et al. [20] proposed UNETR, which combines a Transformer-based encoder with a U-Net-style decoder to produce accurate segmentation results. nnFormer [21] introduced 3D self-attention mechanisms and multi-scale feature learning, demonstrating excellent performance in 3D medical image segmentation tasks. UNETR++ [22] further introduced an efficient paired attention module to model features along both spatial and channel dimensions in parallel, improving segmentation accuracy and computational efficiency, although challenges remain for organs with irregular shapes or blurry boundaries. Yang et al. [23] proposed DTC-SUNETR, effectively integrating the advantages of both Transformers and CNNs. Through the Channel-Spatial Self-Attention (CSSA) module, it enhances the interaction between global and local channel and spatial features, significantly improving the accuracy and efficiency of 3D medical image segmentation. To address the challenge of early diagnosis of Parkinson's disease, [24] proposed an innovative approach based on Vision Transformers (ViT) [25], combined with ElasticNet feature selection and machine learning classifiers, achieving more efficient and accurate diagnostic performance. Hasan Genç et al. [26] proposed a method combining ViT models with ElasticNet feature selection for the automated diagnosis of meniscal tears, achieving 99.9% classification accuracy and 1.2 s inference time, and demonstrating superior performance compared to conventional CNN-based approaches. However, the main obstacle faced by Transformers is the high computational cost of the self-attention mechanism, whose complexity grows quadratically with the increase in image resolution.

With the introduction of the novel Mamba architecture, substantial progress has been made in computer vision tasks due to its outstanding global context modeling capabilities and high computational efficiency [27]. Liu et al. [28] proposed VMamba, designed to achieve visual representation learning with linear time complexity. VMamba integrates selective state space models and introduces a novel 2D selective scan module, enabling the model to adapt to visual data while ensuring computational efficiency. U-Mamba [29] was developed for biomedical image segmentation, combining the strengths of CNNs and SSMs to capture both local features and long-range dependencies in images. Mamba-UNet [30], based on a pure Vision Mamba encoder–decoder structure, uses skip connections to retain spatial information at different scales. Xing et al. [31] proposed SegMamba, a novel 3D medical image segmentation model that maintains excellent inference efficiency while demonstrating strong capabilities in modeling long-range dependencies within voxel data.

Despite these notable advancements achieved by fully supervised deep learning models in medical image segmentation, they heavily rely on large amounts of annotated data. The high cost of obtaining annotations and the tendency for overfitting on small datasets or limited annotations can adversely affect segmentation performance.

2.2. Semi-supervised medical image segmentation

SSL is a training paradigm that combines a small amount of labeled data with a large amount of unlabeled data. Among semi-supervised learning methods in medical image segmentation, two commonly used strategies are consistency regularization [32] and pseudo-labeling [33].

Consistency regularization encourages the model to produce similar outputs for similar inputs, thereby enhancing model robustness and stability. This method typically employs basic data augmentations such as random rotations, scaling, and flipping [34], or more advanced techniques like CutMix [35] to generate diverse inputs and enforce prediction consistency under perturbations. Cross Consistency Training (CCT) [36] proposed a strategy that applies perturbations to the encoder outputs of unlabeled data and uses a shared encoder to enforce consistency among the perturbed outputs. Guided Collaborative Training (GCT) [37] is another common SSL method. During training, GCT uses two segmentation networks with identical structures but different initializations. These networks are collaboratively trained by minimizing a consistency loss between their predictions. The Mean Teacher model [3] introduces two neural networks during training: a student model and a teacher model. The teacher generates pseudo-labels for the student to learn from, with the teacher's parameters updated via an exponential moving average of the student's parameters.

Pseudo-labeling generates labels for unlabeled data using a model trained on labeled samples, enabling further training on the combined dataset. Virtual Adversarial Training (VAT) [38] introduces a virtual adversarial loss to enhance model smoothness around input data points, thereby improving robustness. FixMatch [39] combines strong data augmentations with pseudo-label filtering to further enhance pseudo-label quality and training effectiveness. Built upon the FixMatch framework, UniMatch [40] enhances perturbation strategies by applying strong augmentations at the image level and introducing additional perturbations at the feature level, thereby improving model performance under limited labeled data conditions.

However, pseudo-labels generated by a single model often vary significantly in quality. Thus, generating more robust pseudo-labels through multiple models becomes important. Liu et al. [5] proposed TraCoCo to address inconsistencies in overlapping lesion predictions across different background organs, achieving excellent results in 3D medical image segmentation.

For 3D medical image segmentation, co-training has demonstrated strong capability in leveraging the rich spatial and contextual information inherent in volumetric data. But it still faces challenges such as pseudo-label noise and model confirmation bias. To address these issues, recent co-training approaches have focused on improvements from three main perspectives: data-level, feature-level, and network-level enhancements. The Bidirectional Copy-Paste (BCP) [41] is a data-level approach that enhances the diversity of training samples and the quality of pseudo-supervision by performing region-level bidirectional copy-pasting between labeled and unlabeled images. This method features a simple structure, requires no additional network modules, and can be flexibly integrated into existing segmentation models. Due to potential semantic or stylistic inconsistencies in the synthesized images, BCP is sensitive to the placement and boundary handling of the pasted regions, which may introduce noise into the pseudo labels. At the feature level, the Mutual Consistency Learning (MC-Net+) method [42] builds multiple decoders with slight architectural differences on top of a shared encoder. The model estimates uncertainty regions by measuring discrepancies between decoder outputs, which are then used to guide pseudo-label selection. Furthermore, soft pseudo-labels generated by one decoder are used as training targets for another, establishing cross-branch consistency constraints. At the network level, the Cross Teaching method [43] constructs a CNN branch and a Transformer branch, each focusing on local detail extraction and global semantic modeling, respectively. With their clearly different structures, the two branches perform complementary learning on unlabeled data through cross pseudo-label supervision, effectively leveraging their individual strengths. This heterogeneous architecture not only improves the diversity and reliability of pseudo labels but also enhances the model's capacity to represent structurally complex regions. However, integrating the Transformer architecture undoubtedly increases the number of parameters and the complexity of the model. DC-Net [44] introduces diverse upsampling strategies across different decoders to enhance output diversity, while incorporating the Signed Distance Map as an auxiliary task to guide the model in learning more precise geometric structures, thereby enabling low-entropy and high-accuracy segmentation predictions.

In addition, Semi-Mamba-UNet [45] was the first to apply Mamba to semi-supervised image segmentation tasks with limited labeled data, using two networks to generate pseudo-labels and perform mutual cross-supervision, while also incorporating pixel-level contrastive learning to enhance feature learning capabilities. Nevertheless, this method has not yet been extended to voxel-wise segmentation tasks.

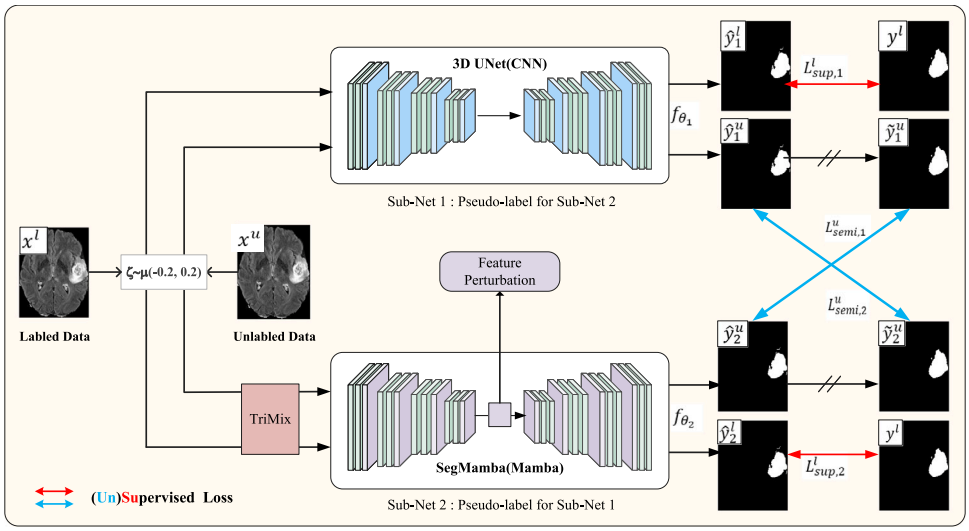

Fig. 2. Overview of the proposed framework for 3D medical image segmentation. The architecture consists of two complementary subnetworks (f_{θ_1} and f_{θ_2}), which supervise each other by generating and utilizing pseudo-labels in a cross-supervised manner. We use the supervised loss L_{sup} and the consistency loss L_{semi} to perform cross-supervision. The subscript i denotes the ith sub-net and the superscript α denotes the labeled data or the unlabeled data. The mark // represents the stop gradient operation.

3. Method

3.1. Co-training segmentation network combining CNN and Mamba

For 3D medical image semi-supervised segmentation, the training set is divided into two subsets: a small labeled dataset $D_L = \{(x_i, y_i)\}_{i=1}^{|D_L|}$ and a large unlabeled dataset $D_U = \{x_i\}_{i=1}^{|D_U|}$, where $|D_L| \ll |D_U|$. Here, $x \in \mathbb{R}^{H\times W\times D}$ is the 3D image, and $y \in \{0, 1, \ldots, S-1\}^{H\times W\times D}$ is the corresponding voxel-wise segmentation label(S represents the total number of classes, binary when $S = 2$).

As shown in Fig. 2, the overall framework of our proposed method is designed to leverage the respective strengths of CNN and Mamba architectures to achieve more accurate 3D medical image segmentation. In this framework, two segmentation networks collaborate with each other: each learner independently generates pseudo-labels that emphasize either local or global structural information, and these pseudo-labels are used to supervise the prediction of the other learner, forming a cross-supervision mechanism. Specifically, 3D UNet and SegMamba are denoted as f_{θ_1} and f_{θ_2}, respectively. The prediction of either network can be treated as pseudo-labels to augment the unlabeled dataset as $(x^u, \tilde{y}^u) \in D_U$, which are then used to train the other network. The powerful global context modeling capability of Mamba helps alleviate CNN's tendency to over-focus on local details, while the local structural feature extraction ability of CNN complements Mamba by enhancing its modeling of fine-grained boundaries and spatial variations. This learning strategy enables the model to effectively capture both local and global semantic information, generating more accurate and robust pseudo-labels, and thereby significantly improving the overall performance of 3D medical image segmentation. During the training phase, the CNN and Mamba branches act as teacher models for each other to facilitate co-training. During the testing or inference phase, only the CNN branch is used for inference.

The optimization of the model is based on the following loss minimization:

$$L(\theta_1, \theta_2) = L_{sup}(D_L, \theta_1, \theta_2) + \lambda L_{sem}(D_U, \theta_1, \theta_2) \tag{1}$$

where θ_1 and θ_2 are parameters of two different networks, L_{sup} is the supervised loss, and L_{sem} is the semi-supervised loss, defined below in (3). λ in (1) is a cosine ramp-up function that controls the trade-off between the losses.

The supervised learning loss is defined as:

$$L_{sup}(D_L, \theta_1, \theta_2) = \sum_{(x,y)\in D_L} L_{ce}(y^l, \hat{y}_1^l) + L_{ce}(y^l, \hat{y}_2^l) + L_{dice}(y^l, \hat{y}_1^l) + L_{dice}(y^l, \hat{y}_2^l) \tag{2}$$

where $L_{ce}(\cdot)$ represents the cross-entropy loss, $L_{dice}(\cdot)$ represents the Dice loss, and $\hat{y}_1^\ell = f_{\theta_1}(x^l + \xi)$ (similar to y_2^l, where $\xi \sim \mu(-0.2, 0.2)$ indicates the noise sampled from the interval $[-0.2, 0.2]$). In Eq. (1), $L_{sem}(\cdot)$ is the semi-supervised loss, which uses the two models' predictions as pseudo-labels. The semi-supervised loss is defined as:

$$L_{sem}(D_U, \theta_1, \theta_2) = -\sum_{x\in D_U}\sum_{\omega\in\Omega} L_{cac}(\hat{y}_1^u(\omega), \hat{y}_2^u(\omega)) + L_{cac}(\hat{y}_2^u(\omega), \hat{y}_1^u(\omega)) \tag{3}$$

where $\hat{y}_1^u(\omega) \in [0, 1]^2$ represents the background $\hat{y}_1^u(\omega)[0]$ and foreground $\hat{y}_2^u(\omega)[1]$ segmentation probabilities at the voxel ($\omega \in \Omega$), which are obtained from the softmax activation function of model 1(model 2 is similar). Ω represents the voxel lattice of the input scan. Note that our proposed $\ell_{\text{cac}}(\cdot)$ replaces the more common MSE loss used in semi-supervised learning methods, and its definition is provided in (9).

3.2. Confidence-aware consistency loss

Co-training based on heterogeneous networks enables two models to learn from different perspectives. However, the training process may suffer from instability due to conflicting predictions between the two networks. In semi-supervised learning, the effectiveness of pseudo-labels plays a crucial role in determining the quality of training on unlabeled data. Therefore, to enhance training stability and improve the reliability of pseudo-labels, we propose a confidence-aware consistency loss. This loss incorporates prediction disagreement detection and class-specific confidence gating, aiming to leverage the structural sensitivity and predictive discrepancy of dual-branch models to guide the network toward more accurate category boundaries and foreground regions.

Specifically, let two sub-networks $f_{\theta 1}$ and $f_{\theta 2}$ generate prediction probabilities for each voxel ω denoted as $\hat{y}_1(\omega)$ and $\hat{y}_2(\omega) \in [0, 1]^2$, where the two dimensions represent the probabilities of the voxel belonging to the background and foreground, respectively. We first determine whether the predicted categories from the two branches

at each voxel are consistent and define a binary variable to indicate prediction discrepancy. If the predicted categories differ, we set $\delta_c = 1$; otherwise, we set $\delta_a = 1 - \delta_c$. Based on this, we design a class-wise confidence-aware pseudo-labeling loss function to select reliable pseudo-labels from the network outputs. For any given prediction a, we only apply supervision when it satisfies the foreground confidence condition $\max(a) > \gamma$ or the background confidence condition $\min(a) < \beta$. First, we define the following loss function as selective confidence loss:

$$\begin{aligned}\ell(a,b) = {} & \mathbb{I}(a > \gamma) \cdot L_{ce}(\mathrm{OneHot}(a), b) \\ & + \mathbb{I}(a \le \beta) \cdot L_{ce}(1 - \mathrm{OneHot}(a), 1 - b)\end{aligned} \tag{4}$$

Here, a and b denote the predicted probability distributions output from different network branches. $\mathrm{OneHot}(a)$ represents the one-hot encoding of the category with the highest predicted probability in a, and $\ell_{\mathrm{ce}}(\cdot)$ refers to the cross-entropy loss function. The indicators $\mathbb{I}(a > \gamma)$ and $\mathbb{I}(a \le \beta)$ represent the confidence conditions for foreground and background supervision, respectively. Their definitions are as follows:

$$\mathbb{I}(a > \gamma) = \begin{cases} 1, & \text{if } \max(a) > \gamma \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

$$\mathbb{I}(a \le \beta) = \begin{cases} 1, & \text{if } \max(a) \le \beta \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

Here, γ and β are hyperparameters used to balance the supervision strength for the foreground and background categories. The positive threshold (γ) and negative threshold (β) were set to 0.65 and 0.3, respectively, following previous studies [5], to ensure stable performance. Based on the aforementioned region partition and confidence gating functions, we further construct two structural pseudo-labeling loss components, organized according to region type.

The first component is the pseudo-label loss for disagreement regions, which focuses on supervising the semantic alignment of the two models in areas with prediction inconsistency. It is defined as:

$$\ell_c = \sum_{\omega \in \hat{\Omega}} \delta_c \cdot \left[\ell(\hat{y}_1(\omega), \hat{y}_2(\omega)) + \ell(\hat{y}_2(\omega), \hat{y}_1(\omega))\right] \tag{7}$$

The second component is the pseudo-label loss for agreement regions, which introduces weaker collaborative guidance when the model predictions are consistent, in order to reduce the risk of confirmation bias. It is formulated as:

$$\ell_a = \sum_{\omega \in \hat{\Omega}} \delta_a \cdot \left[\ell(\hat{y}_1(\omega), \hat{y}_2(\omega)) + \ell(\hat{y}_2(\omega), \hat{y}_1(\omega))\right] \tag{8}$$

To mitigate the potential risk of introducing incorrect pseudo-labels in agreement regions — especially in the early stages of training when the model is not yet well-optimized — we introduce a loss weighting factor $\mu \in (0,1)$ for the agreement term. This factor helps to reduce the interference of such regions during early training. Finally, the complete consistency-aware confidence loss (CAC) is defined as:

$$L_{cac} = \ell_c + \mu \cdot \ell_a \tag{9}$$

where $\mu \in (0,1)$ is a fixed weighting factor controlling the influence of the agreement loss term, which is set to 0.3 in our experiments. The proposed CAC loss serves as a replacement for the commonly used MSE loss in semi-supervised learning, with the goal of maintaining the robustness of MSE against pseudo-label noise. Additionally, it improves training efficiency by selectively enhancing reliable pseudo-labels, which is particularly beneficial in medical image segmentation scenarios where the foreground class is often sparse and occupies a relatively small area.

3.3. Data augmentation

Data augmentation is a crucial technique to enhance model generalization in 3D medical image segmentation tasks. By introducing variability into the training process, the model is exposed to a broader range of examples, enabling it to better adapt to diverse inputs and improve robustness. In our framework, we employ two complementary augmentation strategies: image-space augmentation using TriMix, which operates directly on input images and labels, and feature-space augmentation via Dropout-based perturbation, which introduces variability in the intermediate feature representations.

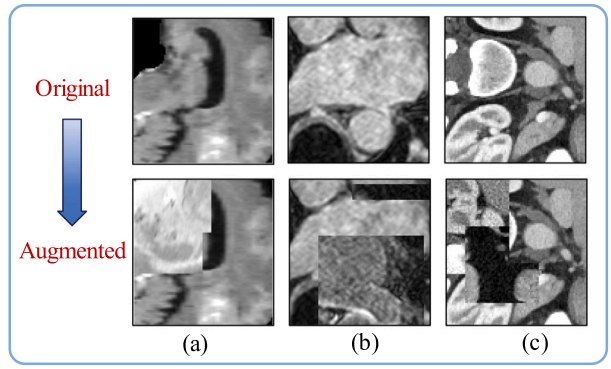

Fig. 3. Comparison of TriMix data augmentation on representative slices from three medical imaging datasets: (a) BraTS2019, (b) LA, (c) Pancreas.

First, to enrich the diversity of spatial structures, we adopt TriMix, an extension of 3D CutMix. This method generates three-valued masks $m_1, m_2, m_3 \in \{0,1\}^{\hat{H}\times\hat{W}\times\hat{D}}$ by sampling random positions and sizes within a cube region from the input $x_i \in \mathbb{R}^{H\times W\times D}$. For each mini-batch, the masked regions are simultaneously applied to unlabeled inputs and their pseudo-labels, providing diverse spatial cues for model training. Formally, given a group of unlabeled 3D volumes x_{ij}, corresponding labels y_{ij}, and a randomly generated mask $m \in \{0,1\}^{\hat{H}\times\hat{W}\times\hat{D}}$, the transformed input is defined as:

$$\begin{aligned} v_{ij}^{m_1} &= (1 - m_1) \odot x_i + m_1 \odot x_j, \\ v_{ijk}^{m_2} &= (1 - m_2) \odot v_{ij}^{m_1} + m_2 \odot x_k, \\ y_{2,ij} &= (1 - m_1) \odot y_{2,i} + m_1 \odot y_{2,j}, \\ y_{2,ijk} &= (1 - m_2) \odot y_{2,ij} + m_2 \odot y_{2,k}, \end{aligned} \tag{10}$$

where i, j, k represent different samples within a batch ($i \ne j \ne k$) to ensure each contributes unique portions to the final 3D volume, and $\odot$ denotes element-wise multiplication. In TriMix, the cutout region is not fixed but sampled dynamically at each iteration. Specifically, the mask size is determined by a bounding box generated according to a Beta(4,4) distribution, following the formulation in CutMix. In our implementation, TriMix augmentation is applied to every training batch. This strategy exposes the model to more diverse mixed 3D inputs and labels, improving robustness in handling unlabeled medical images. Furthermore, we provide a visualization that illustrates the comparison between the original inputs and the TriMix-augmented samples, as shown in Fig. 3.

In addition to spatial augmentation in the image domain, we also introduce feature-space augmentation by applying dropout-based perturbation to the intermediate feature maps. Mathematically, for an intermediate feature map F, the perturbation is defined as:

$$\hat{F} = \mathcal{P}(F, p) \tag{11}$$

where $\mathcal{P}$ denotes feature perturbation, we adopt a simplest form of feature perturbation in our method, which is dropout, p is the dropout rate of 0.5. There are some other options available, such as uniform noise and VAT [38]. We follow [40] to set the strategy. While dropout is typically regarded as a regularization technique, in this work it serves as a form of feature-level augmentation by randomly perturbing the network's latent representations. Specifically, we apply dropout

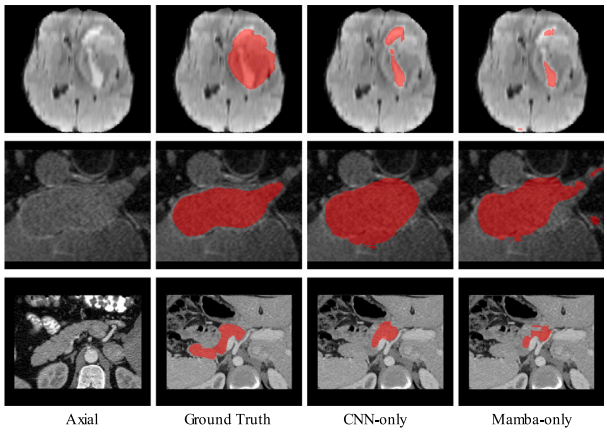

Fig. 4. Representative axial slices from three datasets and corresponding segmentation results with 10% labeled data. Predictions from CNN-only and SegMamba are compared with ground truth, illustrating the limitations of fully supervised models under scarce annotations.

to the intermediate feature maps of the SegMamba network during training. For each mini-batch, dropout randomly sets a proportion of feature activations to zero, simulating different network responses and encouraging the model to learn more robust and noise-tolerant representations. The perturbed features are then used for subsequent segmentation predictions.

Together, these two strategies enrich both the input space and feature space. TriMix improves exposure to diverse anatomical patterns, while Dropout3D enforces robust representation learning. Their combination significantly enhances the stability and generalization of our semi-supervised segmentation framework.

4. Experiments

4.1. Datasets

To evaluate the effectiveness of our method, we conducted experiments on three public datasets: Brain Tumor Segmentation 2019 (BraTS19) [46], Left Atrium (LA) [47], and Pancreas-CT (Pancreas) [48].

BraTS2019 Dataset. This dataset contains preoperative MRI scans of 335 brain tumor patients, including 259 cases of high-grade gliomas (HGG) and 76 cases of low-grade gliomas (LGG). Each case includes four different MRI modalities: T1-weighted (T1), contrast-enhanced T1-weighted (T1-ce), T2-weighted (T2), and T2 fluid-attenuated inversion recovery (FLAIR). All images have a resolution of $240 \times 240 \times 155$ and are co-registered to the same spatial resolution ($1.0 \times 1.0 \times 1.0$ mm^3). Since FLAIR is highly effective in characterizing malignant tumors [8], and most previous semi-supervised segmentation studies have also relied solely on the FLAIR sequence [49], we likewise employ FLAIR and adopt the same data partitioning and preprocessing procedures to ensure fair comparison [50]. In addition, related clinical studies have investigated the advantages of FLAIR imaging in neurosurgery, highlighting its critical role in brain tumor surgery [51].

LA Dataset. This dataset contains 154 MRI scan samples from 60 patients diagnosed with atrial fibrillation, among which 100 scans are annotated. The scans have an isotropic resolution of $0.625 \times 0.625 \times 0.625$ mm^3, and the segmentations were manually delineated by expert radiologists.

Pancreas Dataset. The Pancreas dataset consists of 82 3D contrast-enhanced CT scans collected from 53 male and 27 female patients at the NIH Clinical Center. Each slice has a resolution of 512×512 pixels, with a slice thickness ranging from 1.5 to 2.5 mm. During preprocessing, we follow the method described in [52], clipping voxel values to the range of [−125, 275] Hounsfield Units and resampling the data to an isotropic resolution of $1.0 \times 1.0 \times 1.0$ mm^3.

Table 1
Dataset splits for training and validation in our experiments.

Dataset	Total	Val	Train	Training data partition		
				10%	20%	40%
BraTS19	335	60	250	25	50	100
LA	100	20	80	8	16	32
Pancreas	82	20	60	6	12	–

Fig. 4 presents representative axial slices from three datasets along with the corresponding segmentation results under the 10% labeled setting. As can be observed, the fully supervised methods (CNN-only and Mamba-only) result in coarse and incomplete segmentations, with significant deviations from the ground truth, often leading to either over-segmentation or under-segmentation. These examples not only serve as dataset illustrations but also highlight the strong reliance of fully supervised approaches on sufficient annotations, underscoring the need for semi-supervised methods.

As shown in Table 1, we present the dataset splits for training and validation in our experiments. For each dataset, we have specified the total number of cases, as well as the number of cases designated for training and validation. Following previous work [50,52], we adopt the same data split and preprocessing procedure, ensuring fair comparison with other methods.

4.2. Implementation details

Training and Testing Configurations. The method in this paper is implemented using PyTorch and trained on an NVIDIA Quadro RTX6000 GPU with 24 GB of memory on Ubuntu 16.04. The initial learning rate is set to 5e–2. The maximum number of iterations for the BraTS19 dataset is set to 30,000, while for the LA and Pancreas datasets, it is set to 15,000. The model is trained using mini-batch stochastic gradient descent (SGD) with momentum, where the momentum is fixed at 0.9 and weight decay is set to 0.0005. Based on the settings in [42], the LA dataset is cropped to $112 \times 112 \times 80$, while the BraTS19 and Pancreas datasets are cropped to $99 \times 96 \times 96$. During the testing phase, a network $f_{\theta 1}$ is used for result prediction, and the final segmentation results are obtained using a sliding window strategy. For the BraTS2019 and Pancreas datasets, the window stride is $16 \times 16 \times 16$, while for the LA dataset, the stride is $18 \times 18 \times 41$. To further illustrate the training dynamics of our method, we plot the variations of the training loss and the Dice score with respect to the number of iterations. The evolution of Dice accuracy and loss during training on the BraTS2019 dataset (with 10% labeled data) is shown in Fig. 5, both Model 1 and Model 2 exhibit a consistent decrease in loss accompanied by a steady increase in Dice score, which indicates stable convergence and reliable optimization during training.

Compared Methods. Several mainstream semi-supervised image segmentation approaches and supervised baseline are compared on BraTS2019, LA, and Pancreas, including: UA-MT [50], cross pseudo supervision(CPS) [6], interpolation consistency training(ICT) [53], UniMatch [40], Translation Consistent Co-training(TraCoCo) [5], shapeaware semi-supervised segmentation network(SASSNet) [54], uncertainty rectified pyramid consistency(URPC) [55], MccauSSL [56], 3DUnet [17] and SegMamba [31]. In the case of the BraTS19 dataset, the results of the current methods are based on our re-implementation. The LA and Pancreas datasets are commonly used benchmarks for evaluating semi-supervised 3D medical segmentation methods; therefore, the results of some methods are taken from their original papers.

Evaluation Metrics. We adopt four evaluation metrics: Dice coefficient, Jaccard similarity coefficient (Jaccard), 95%Hausdorff Distance (HD95), and Average Surface Distance (ASD). The Dice coefficient assesses the similarity between segmentation results while the Jaccard

Table 2
Comparison results on BraTS2019 dataset.

Model	Training set		Metrics			
	Labeled	Unlabeled	Dice (%)↑	Jaccard (%)↑	ASD↓	HD95↓
UA-MT [50]	25 (10%)	225 (90%)	84.05	74.33	2.26	10.26
CPS [6]			84.20	74.28	2.76	10.72
ICT [53]			83.71	73.62	2.65	12.09
UniMatch [40]			85.03	75.91	2.42	9.50
TraCoCo [5]			85.19	**76.10**	2.16	9.32
Ours			**85.39**	76.06	**1.75**	**7.82**
UA-MT [50]	50 (20%)	200 (80%)	84.66	75.18	2.38	10.53
CPS [6]			84.07	75.33	2.20	8.69
ICT [53]			84.61	74.35	2.34	9.59
UniMatch [40]			85.48	76.07	2.76	10.03
TraCoCo [5]			85.98	**76.37**	2.53	9.83
Ours			**86.24**	76.75	**2.11**	**8.42**
UA-MT [50]	100 (40%)	150 (60%)	85.88	76.02	2.27	9.20
CPS [6]			86.02	75.87	2.01	10.26
ICT [53]			86.22	76.99	1.94	7.48
UniMatch [40]			86.03	76.91	2.42	9.50
TraCoCo [5]			86.61	77.39	1.73	6.98
Ours			**86.97**	**77.44**	**1.42**	**5.52**

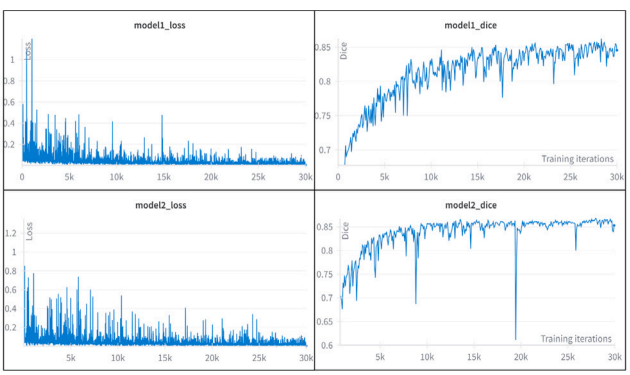

Fig. 5. Loss and dice accuracy on BraTS2019 with 10% labels.

coefficient measures the consistency of shared features between samples. The HD95 evaluates the distance between two point sets, and the ASD measures the average surface distance between the ground truth and the segmentation results. Let X be the model's prediction, and Y be the ground truth. $|X|$ and $|Y|$ represent the voxel counts of X and Y, respectively. N denotes the number of pixels on the segmentation boundary, and $d(x, y)$ represents the distance between voxel $x \in X$ and the closest voxel $y \in Y$. The evaluation metrics are defined as:

$$Dice(X,Y) = \frac{2|X \cap Y|}{|X| + |Y|} \times 100\% \tag{12}$$

$$Jaccard(X,Y) = \frac{|X \cap Y|}{|X \cup Y|} \times 100\% \tag{13}$$

$$ASD(X,Y) = \frac{1}{|X|} \sum_{x \in X} \min_{y \in Y} d(x,y) \tag{14}$$

$$HD95(X,Y) = 0.95 \times \max \left\{ \max_{x \in X} \min_{y \in Y} d(x,y), \max_{y \in Y} \min_{x \in X} d(x,y) \right\} \tag{15}$$

To ensure fair comparison, all baseline methods were trained and evaluated under the same experimental conditions, including identical labeled/unlabeled splits(as detailed in Table 1), cropping sizes, and augmentation strategies. Thus, performance differences reflect only the effectiveness of the model designs rather than discrepancies in training setup.

4.3. Comparison and analysis with semi-supervised methods

The comparison results on the BraTS2019 dataset are shown in Table 2, which presents the performance comparison of different methods under various annotation ratios (10%, 20%, 40%). When only 10% of the data is annotated, the proposed method outperforms other methods in terms of Dice, ASD, and HD95 metrics, achieving values of 85.39%, 1.75, and 7.82, respectively. The Jaccard score also reaches 76.06%, which is close to the optimal result. Although TraCoCo achieves a slightly higher Jaccard score, it performs worse on ASD and HD95, indicating reduced robustness in boundary delineation and outlier control. With 20% annotated data, the performance further improves, with Dice reaching 86.24%, ASD reduced to 2.11, and HD95 to 8.42. For the 100-label partition protocol, our approach is 0.36% and 1.46 better than the TraCoCo in terms of Dice and HD95, respectively. These results surpass all comparison methods, particularly in terms of boundary smoothness and outlier handling, highlighting the fine-grained segmentation capability of our framework.

Fig. 6 presents qualitative comparisons of brain tumor segmentation results obtained with 10% labeled data. The first, second, and third rows correspond to axial, sagittal, and coronal views, respectively. Fig. 7 further shows segmentation results using 20% labeled data on the BraTS2019 dataset. It can be observed that, the proposed method produces segmentation outputs that are visually closer to the ground truth compared with other methods.

The segmentation comparison results on the LA dataset are shown in Table 3. As shown, our proposed method achieves superior performance compared to other methods, further demonstrating its effectiveness and robustness for left atrium segmentation. For instance, our approach increases Dice by 1.51% and 0.62%, compared with UniMatch and TraCoCo under the 8-sample partition protocol. Similarly, we increase Jaccard by 0.57% over TraCoCo for the 16-sample partition protocol. At 40% labeled data, our method achieves the best performance, with Dice at 92.07%, Jaccard at 86.09%, and further reductions in ASD and HD95 to 1.29 and 4.13, respectively. Particularly, our method significantly outperforms the second-best approach, underscoring its superior capability in segmenting structurally complex regions such as atrial boundaries.

Table 4 presents the performance comparison on the Pancreas dataset under different annotation ratios. Despite the pancreas being a small organ with complex morphology and ambiguous boundaries, our method consistently achieves superior results. With only 10% of labeled data, it obtains Dice and Jaccard scores of 78.45% and 65.75%, both outperforming other methods. In terms of ASD and HD95, it achieves scores of 2.95 and 10.15. At the 20% annotation ratio, our method continues to maintain top performance across all metrics. The Dice score increases to 82.45%, Jaccard reaches 70.30%, respectively. Our approach yields the best performance in HD95, where it outperforms TraCoCo and CPS by 0.31 and 1.38 voxels.

Table 3
Comparison results on LA dataset.

Model	Training set		Metrics			
	Labeled	Unlabeled	Dice (%)↑	Jaccard (%)↑	ASD↓	HD95↓
UA-MT [50]	8 (10%)	72 (90%)	86.15	75.98	4.57	18.50
CPS [6]			87.43	78.59	3.70	13.12
ICT [53]			85.26	74.92	2.88	9.55
UniMatch [40]			88.23	78.90	2.72	8.10
TraCoCo [5]			89.12	80.63	2.24	6.85
Ours			**89.74**	**81.62**	**1.98**	**6.70**
UA-MT [50]	16 (20%)	64 (80%)	88.63	79.85	2.30	8.20
CPS [6]			89.49	81.20	2.10	7.08
ICT [53]			88.51	81.26	1.99	6.91
UniMatch [40]			90.86	83.42	1.93	6.01
TraCoCo [5]			90.78	83.35	1.76	**5.42**
Ours			**91.13**	**83.92**	**1.82**	5.65
UA-MT [50]	32 (40%)	48 (60%)	90.03	82.66	1.61	7.05
CPS [6]			91.20	83.29	1.63	4.97
ICT [53]			90.91	83.51	1.62	5.70
UniMatch [40]			91.44	84.27	1.50	4.95
TraCoCo [5]			91.78	85.04	1.43	4.20
Ours			**92.07**	**86.09**	**1.29**	**4.13**

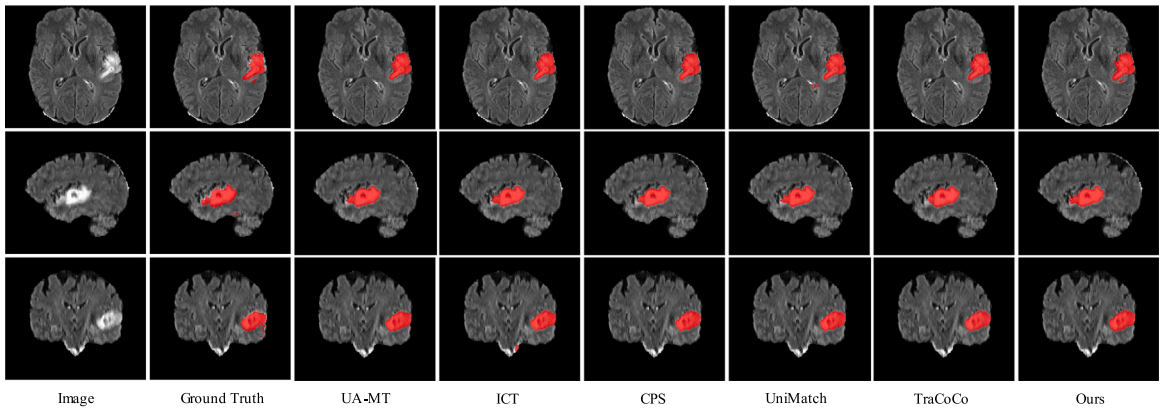

Fig. 6. Comparative visualization results with other semi-supervised methods on the BraTS2019 dataset using 10% labeled data: (a) Original image; (b) Ground truth image; (c) UAMT; (d) ICT; (e) CPS; (f) UniMatch; (g) TraCoCo (i) Our method.

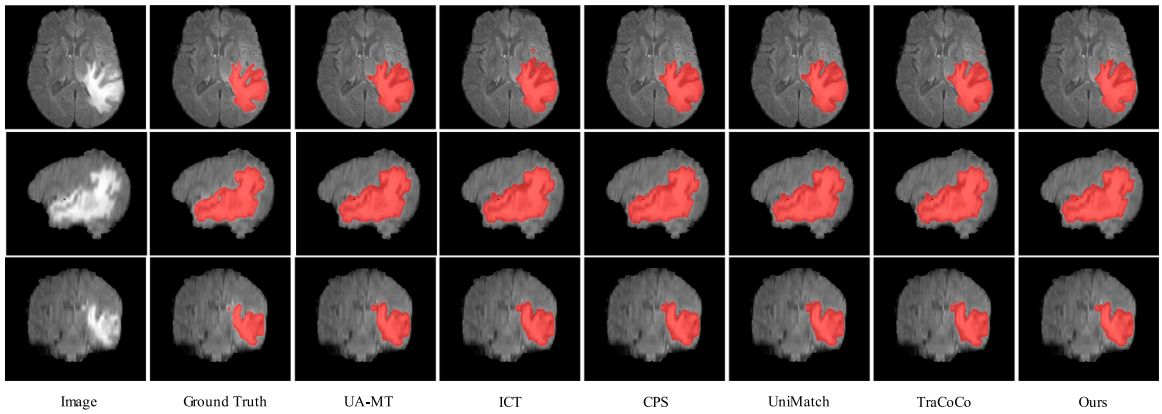

Fig. 7. Comparative visualization results with other semi-supervised methods on the BraTS2019 dataset using 20% labeled data: (a) Original image; (b) Ground truth image; (c) UAMT; (d) ICT; (e) CPS; (f) UniMatch; (g) TraCoCo (i) Our method.

Table 4
Comparison results on the Pancreas dataset.

Model	Training set		Metrics			
	Labeled	Unlabeled	Dice (%)↑	Jaccard (%)↑	ASD↓	HD95↓
UA-MT [50]	6 (10%)	56 (90%)	66.42	52.13	3.07	17.15
CPS [6]			69.13	54.43	2.82	14.63
SASSNet [54]			68.97	54.29	1.96	18.83
URPC [55]			73.53	59.44	7.85	22.57
MccauSSL [56]			72.89	58.06	4.37	14.19
UniMatch [40]			76.82	62.43	3.72	13.23
TraCoCo [5]			78.32	65.23	**2.92**	10.23
Ours			**78.45**	**65.75**	2.95	**10.15**
UA-MT [50]	12 (20%)	50 (80%)	76.23	62.73	2.42	10.93
CPS [6]			78.33	64.83	2.22	8.43
SASSNet [54]			76.39	63.17	**1.42**	11.06
URPC [55]			80.02	67.30	1.98	8.51
MccauSSL [56]			80.92	68.26	1.53	8.11
UniMatch [40]			82.33	70.32	1.61	7.13
TraCoCo [5]			82.30	70.32	1.91	7.36
Ours			**82.45**	**70.30**	1.95	**7.05**

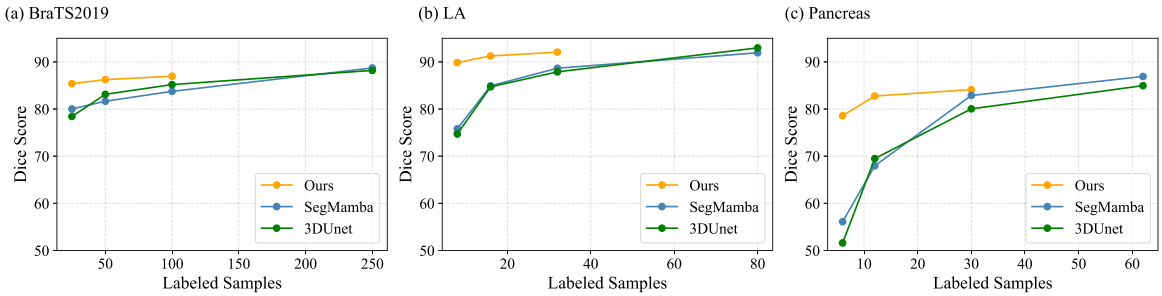

Fig. 8. Dice comparison between ours and fully supervised methods (a) BraTS2019; (b) LA; (c) Pancreas.

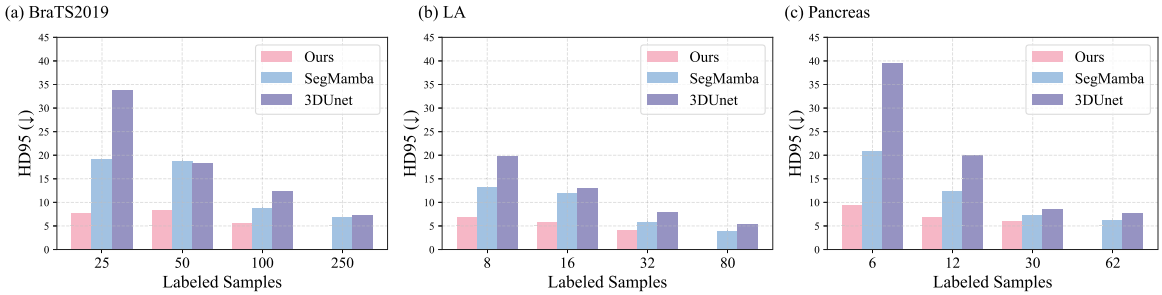

Fig. 9. HD95 comparison between ours and fully supervised methods (a) BraTS2019; (b) LA; (c) Pancreas.

In addition to the quantitative analysis presented above, we further conduct a qualitative evaluation. Both the metric-based results and the visual comparisons consistently demonstrate that our proposed model produces segmentation outputs highly aligned with the ground truth across all datasets, particularly excelling in capturing fine lesion boundaries and complex anatomical structures. On the BraTS dataset, our method shows a stronger ability to reconstruct the overall morphology of tumor lesions, especially in delineating subtle boundary regions where it significantly outperforms other approaches. As shown in Fig. 4, fully supervised methods tend to yield incomplete tumor segmentation due to the limited number of annotated samples. Among semi-supervised approaches, UA-MT and ICT suffer from blurred or missing boundary details, while CPS performs better in some regions but still struggles with boundary clarity and completeness. Although UniMatch and TraCoCo demonstrate improvements in boundary refinement, their performance remains limited when handling highly complex structures. On the LA and Pancreas datasets, our method also demonstrates clear advantages, exhibiting high sensitivity to lesion details and a strong capability for fine-grained modeling. These improvements are largely attributed to the integration of CNN and Mamba, which enhances the synergy between global and local feature modeling. Furthermore, the proposed CAC loss improves pseudo-label accuracy, and the incorporation of data augmentation further strengthens generalization. Collectively, these components enable our framework to achieve superior robustness and generalization across diverse segmentation tasks, providing a solid foundation for precise medical image segmentation in clinical applications.

4.4. Comparison and analysis with fully supervised methods

To validate the effectiveness of the proposed semi-supervised method, we performed a systematic comparison with fully supervised methods, 3DUnet and SegMamba, on the BraTS2019, LA, and Pancreas datasets. The experimental results are shown in Figs. 8 and 9. From the trend of the Dice scores, it is evident that, under conditions with limited annotated samples, our method achieves superior segmentation performance compared to fully supervised methods across all three datasets. Notably, the performance improvement is more significant when the number of annotated samples is smaller. For example, in the LA dataset,

Table 5
Ablation study of the proposed framework on BraTS2019 and LA Datasets. The upper part compares different architecture variants, and the lower part presents the effect of major components. Including CAC loss and data augmentation (Aug.) strategy.

Setting	BraTS2019				LA			
	Dice (%) ↑	Jaccard (%) ↑	ASD ↓	HD95 ↓	Dice (%) ↑	Jaccard (%) ↑	ASD ↓	HD95 ↓
CNN&Transformer	83.98	72.77	3.28	12.64	87.21	77.08	4.22	13.04
CNN&Mamba (proposed)	**84.67**	**74.83**	**2.13**	**12.12**	**87.96**	**79.01**	**2.33**	**8.12**
CNN&Mamba+CAC Loss	85.07	75.75	2.00	10.69	88.95	80.74	2.18	7.92
CNN&Mamba+Data Augmentation	85.00	75.14	1.88	9.25	89.03	80.90	2.24	7.77
CNN&Mamba+CAC Loss+Aug.	**85.39**	**76.06**	**1.75**	**7.82**	**89.74**	**81.62**	**1.98**	**6.70**

when only 8 annotated samples are used, our method improves Dice score by 15.05% compared to 3DUnet and by 14.05% compared to SegMamba. In the Pancreas dataset, when the number of annotated samples is 6, the Dice score improves by 26.9% over 3DUnet and by 22.48% over SegMamba. In the BraTS2019 dataset, with 25 annotated samples, the Dice score improves by 6.96% over 3DUnet and by 5.37% over SegMamba. These results clearly demonstrate that our method can fully leverage unlabeled data, significantly enhancing the model's learning capability and segmentation accuracy in environments with limited labeled data. As the number of labeled samples increases, the performance of all methods improves, but our method still maintains a certain advantage. On the LA dataset, with 16 and 32 labeled samples, our method improves by approximately 6% and 4%, respectively. On the BraTS2019 dataset, with 50 and 100 samples, the improvements are approximately 4% and 3%, respectively. This indicates that even in scenarios with moderate labeling, the semi-supervised strategy can continue to provide performance gains.

From the comparison of the HD95 metric, it can be further observed that our method also outperforms the fully supervised methods in terms of boundary accuracy. With fewer labeled samples, the HD95 metric is notably lower than the comparison methods, especially in the Pancreas dataset, where the reduction in HD95 is most significant under the condition of 6 labeled samples. This suggests that our method can better capture the boundary details of target structures while maintaining overall segmentation accuracy.

4.5. Ablation study

To further evaluate the effectiveness of each component in our proposed method, we conducted a series of ablation experiments in this section.

4.5.1. Divergence comparison

First, we conducted a divergence analysis between the Co-Training and Mean Teacher frameworks on the BraTS19 dataset. The overlap ratio and Manhattan Distance (also referred to as L1 distance) can be used to evaluate the consistency of outputs from multiple models [5], and their calculation methods are shown in the following formulas.

$$\text{Overlap Ratio} = \frac{\left|\{v \in V \mid f_1(v) = f_2(v)\}\right|}{|V|} \tag{16}$$

$$\text{Manhattan Distance} = \sum_{v \in V} \left|f_1(v) - f_2(v)\right| \tag{17}$$

where V is the set of all voxels, and $|V|$ is the total number of voxels. $\left|\{v \in V \mid f_1(v) = f_2(v)\}\right|$ is the number of voxels with same segmentation results. $f_1(v)$ and $f_2(v)$ denote the outputs of the two models for voxel v. A lower overlap ratio and a larger Manhattan Distance indicate that the two models learn more complementarily and exhibit greater diversity.

As shown in Fig. 10, we compared the similarity of pseudo-labels in the unlabeled portion of the BraTS19 dataset between the Co-Training and Mean Teacher frameworks. These measurements were averaged over each training epoch and calculated at different stages of the training process. Specifically, we computed the voxel-wise overlap ratio

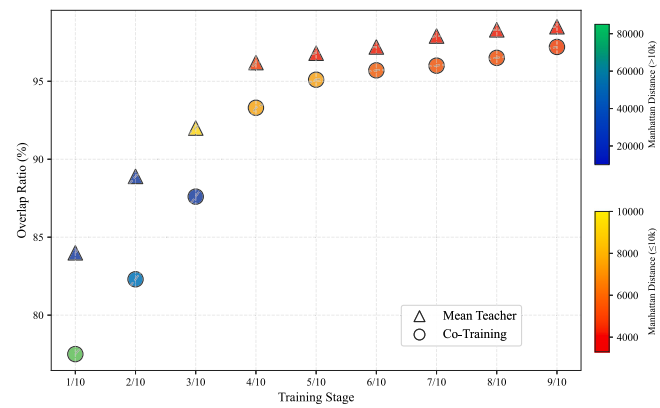

Fig. 10. Comparison of overlap ratio and Manhattan distance across training stages for co-training and Mean Teacher. Circles represent Co-Training results, triangles represent Mean Teacher results. Colors encode Manhattan distance values, with a continuous colormap from high (green/blue) to low (yellow/red).

and the voxel-wise Manhattan Distance of the foreground segmentation results from the two networks. It was observed that, throughout the training process, Co-Training produced a slightly lower overlap ratio and a noticeably larger voxel-wise Manhattan Distance compared with Mean Teacher, indicating greater diversity. In semi-supervised learning, a certain degree of diversity between two models is generally desirable, as it enables them to complement each other during mutual supervision and capture more comprehensive information. Therefore, Co-Training constitutes a more suitable approach for 3D medical image segmentation tasks.

4.5.2. Ablation study of components

Our method is based on two distinct types of network architectures: the traditional convolutional network 3D-UNet and the segmentation network SegMamba built upon the Mamba mechanism. The aim is to combine the local feature extraction strength of CNNs with Mamba's capability for global dependency modeling. Although both networks adopt a U-shaped structure, they represent different learning paradigms. Given that Transformers are also effective at capturing global context owing to their wide receptive fields, we further replaced SegMamba with UNETR to construct a CNN–Transformer architecture. Comparative experiments on the BraTS2019 and LA datasets were conducted to assess the effectiveness of different combinations under cross-supervision. As shown in Table 5, the performance comparison across different combinations on the BraTS2019 and LA datasets, using 10% of the labeled data, demonstrates that the CNN–Mamba co-training architecture consistently outperforms the CNN–Transformer combination under identical annotation conditions. Specifically, Dice scores on BraTS2019 and LA improved by 0.69% and 0.75%, respectively, along with superior results in Jaccard, ASD, and HD95. These results highlight the advantages of the Mamba architecture in feature modeling and global context capture.

Table 6
Loss ablation study on BraTS2019 using 10% of labeled data.

L_{sup}	L_{semi}			Metrics			
	L_{kl}	L_{mse}	L_{cac}	Dice↑	Jaccard↑	ASD↓	HD95↓
✓	✓			84.95	75.46	1.90	9.17
✓		✓		84.90	75.09	1.77	10.16
✓			✓	85.39	76.06	1.75	7.82

We further analyze the impact of two major components in our proposed framework: the CAC Loss and data augmentation strategy. These components were added on top of the CNN&Mamba architecture. Specifically, we replaced the CAC loss with a simple CE loss, applied the data augmentation method, and also evaluated using only the CAC loss without data augmentation. The lower part of Table 5 presents the results of using these components individually and in combination. As shown in Table 5, the full configuration with all three modules achieves the best performance on key metrics such as Dice and HD95. Specifically, when retaining data augmentation in the three components but removing the CAC loss and replacing it with the standard cross-entropy loss, the Dice score on the BraTS2019 dataset decreases by 0.39%, and HD95 increases by 1.43. On the LA dataset, Dice decreases by 0.71%, and HD95 increases by 1.07. When data augmentation is removed, the Dice scores on the BraTS2019 and LA datasets decrease by 0.32% and 0.79%, respectively, while HD95 increases by 2.87 and 1.22, respectively.

4.5.3. Ablation study of loss functions

Finally, we further verify the effectiveness of the proposed CAC loss L_{cac} by conducting an ablation study in which it is replaced in Eq. (3) with the KL divergence loss L_{kl} and the MSE loss L_{mse}, respectively. In all configurations, the supervised segmentation loss L_{sup} remains unchanged. Table 6 presents the experimental results on the BraTS2019 dataset using 10% of labeled data. When L_{kl} is used to form the semi-supervised loss, the model achieves a Dice score of 84.95% and a Jaccard index of 75.46%. Replacing L_{kl} with L_{mse} slightly improves ASD performance but changes HD95 from 9.17 to 10.16, indicating that L_{mse} promotes local smoothness but is less effective in controlling large outlier distances. In contrast, employing the proposed L_{cac} leads to consistent improvements across all metrics, achieving the highest Dice and Jaccard scores while reducing ASD and HD95 to 1.75 and 7.82, respectively. These gains demonstrate the ability of CAC loss to better align fine-grained spatial boundaries, providing more reliable and precise segmentation performance than conventional losses. In particular, traditional loss functions tend to produce unstable results in complex backgrounds or for small organs, leading to lower Dice scores and increased HD95. By comparison, our method improves consistency learning under limited annotation conditions through the dynamic weighting mechanism of L_{cac}, thereby enhancing overall performance.

4.5.4. Ablation study of hyper-parameters

As described in Section 3.2, the hyperparameter μ in Eq. (9) controls the trade-off between the two components of the CAC loss, which is crucial for the effectiveness of the loss function. Therefore, we analyze the impact of different μ values on the overall segmentation performance of the model. The experimental results are presented in Fig. 11, based on which we set μ to 0.3.

These experiments provide a comprehensive understanding of the contribution of each component in our model, and the results indicate that integrating all components is crucial for achieving robust 3D medical image segmentation.

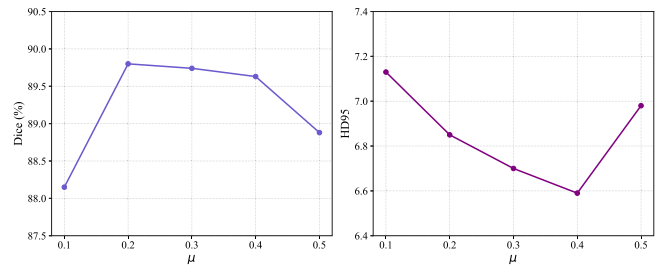

Fig. 11. The segmentation performance of our approach with different values of μ under 10% partition protocol on the LA dataset.

Table 7
Computational analysis on the BraTS2019 Dataset at the ratio of 10%.

Method	Params (M)	FLOPs (G)	Test/case
3DUnet	5.88	122.28	0.48 s
UNETR	92.62	165.07	0.57 s
SegMamba	28.87	334.78	1.15 s
UA-MT	11.77	144.56	0.53 s
ICT	11.77	183.42	0.49 s
TraCoCo	11.76	122.82	0.55 s
Ours	**36.43**	**229.38**	**0.55s**

5. Discussion

5.1. Computational analysis

As shown in Table 7, we compare the proposed method with existing 3D medical image segmentation approaches in terms of computational cost and training efficiency on the BraTS2019 dataset with 10% labeled data. Complexity is measured during the training phase. The evaluation includes the number of parameters, floating-point operations (FLOPs), and inference time. The compared methods can be broadly categorized into two groups: fully supervised and semi-supervised approaches. From the results, it can be observed that Transformer-based models (e.g., UNETR) typically require a large number of parameters and exhibit high computational complexity, which significantly increases hardware demands and limits their applicability in resource-constrained scenarios. In contrast, conventional CNN-based models (e.g., 3D U-Net) contain fewer parameters and lower computational complexity but struggle to achieve the same level of accuracy as Transformer models when dealing with complex lesion regions. Mamba-based models (e.g., SegMamba) reduce the number of parameters compared to Transformer architectures but still suffer from higher computational complexity and longer inference times.

Our proposed method achieves competitive segmentation performance under limited annotation settings while striking a favorable balance among parameter size, inference time, and computational cost. Specifically, the model contains 36.43M parameters, an inference time of 0.55 s, and a computational cost of 229.38 GFLOPs. Although the FLOPs are relatively high, the model substantially reduces parameter size compared to Transformer-based architectures while improving segmentation accuracy and avoiding excessive inference delays. These results highlight the practical feasibility of our method, particularly in resource-constrained environments, demonstrating strong deployment potential and real-world applicability. Overall, this validates our method as an efficient and robust solution for 3D medical image segmentation in semi-supervised scenarios.

5.2. Limitations and future work

Although our model effectively reduces the reliance on annotated data and achieves excellent segmentation accuracy, there are still several areas for improvement. First, while our proposed architecture integrates local and global features through CNN and Mamba, certain critical spatial and contextual information may require more fine-grained attention. A promising direction would be to design a cross-model feature fusion module combined with advanced attention mechanisms to further enhance the ability to capture subtle details. Second, in clinical applications, the requirement for rapid and accurate medical image processing is particularly critical. Our method involves model-level integration, which inevitably increases complexity and may limit scalability in real-world clinical settings. Future work could therefore focus on improving the real-time efficiency of the proposed framework. Exploring model compression techniques, such as knowledge distillation and pruning, may enable the development of lightweight variants suitable for deployment on edge devices, thereby significantly accelerating processing speed while maintaining segmentation accuracy.

6. Conclusion

This paper proposes a semi-supervised 3D medical image segmentation framework based on collaborative training between CNN and Mamba architectures. The method is motivated by the complementary strengths of the two models: CNNs are effective at capturing local spatial features, while Mamba excels at modeling long-range dependencies and sequential patterns efficiently. By jointly training both branches, the network benefits from their synergistic capabilities. To further enhance generalization and alleviate the training bias caused by noisy or low-quality pseudo-labels in traditional approaches, we introduce a CAC loss. This mechanism incorporates prediction consistency analysis and class-specific confidence gating to apply structured pseudo-supervision separately to agreement and disagreement regions, thereby improving the model's discriminative ability and robustness on unlabeled data. In addition, to increase the model's adaptability to structural variations and spatial transformations, we adopt an improved 3D CutMix augmentation strategy and integrate feature perturbation in intermediate layers. These components help mitigate overfitting and further strengthen the model's generalization capability, particularly in scenarios with limited annotations. Experiments were conducted on BraTS2019, LA, and Pancreas. The results demonstrate that the proposed model achieves competitive performance across multiple evaluation metrics, including Dice, Jaccard, ASD, and HD95. In future work, we will explore more efficient semi-supervised strategies to improve segmentation performance and enhance clinical applicability.

CRediT authorship contribution statement

Yun Jiang: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis. **Pengyu Chen:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bingxi Liu:** Software, Investigation, Formal analysis. **Miaofeng Lu:** Visualization, Formal analysis. **Longgang Yang:** Validation. **Yuhang Li:** Software. **Jinliang Su:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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